Application: DemoQA Book Store Application
Module: Login
URL: https://demoqa.com/login
Document ID: TP-DEMOQA-LOGIN-001
Version: 1.0
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Date: September 7, 2026

SOFTWARE TEST PLAN

1. Introduction

The purpose of this test plan is to verify that the DemoQA Login module works correctly, securely, and consistently across supported environments.

Testing will cover authentication, input validation, error handling, session behavior, usability, compatibility, negative scenarios, and basic performance.

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4. Objectives

The objectives are to:

- Verify valid users can log in successfully.
- Verify invalid credentials are rejected.
- Validate username and password fields.
- Verify appropriate error/validation messages.
- Verify password masking.
- Verify session creation and logout.
- Prevent unauthorized access.
- Test negative and boundary inputs.
- Verify browser compatibility.
- Identify and report defects before release.
- Verify fixes through retesting and regression testing.

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5. Scope

5.1 Features Included

- Login page/UI
- Username field
- Password field
- Login button
- Required-field validation
- Valid authentication
- Invalid authentication
- Error messages
- Password masking

- Input handling
- Boundary and special-character testing
- Session/login state
- Logout
- Unauthorized/direct URL access
- Back/refresh behavior after logout
- New User navigation
- Browser compatibility
- Basic performance
- Negative testing
- Regression testing

5.2 Features Excluded

- New User registration validation
- User profile functionality
- Book management
- Book purchasing
- Complete application business logic
- Production penetration testing
- Features outside the Login module

Example:
Testing whether a user can successfully log in is **in scope**. Testing whether the user can add a book after login is outside the Login module's primary scope.

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6. Software Risk Issues

These are risks **within the software** that may affect users or business.

Risk	Impact	Testing Response
Valid user cannot log in	Critical	High-priority functional testing
Invalid user gets access	Critical	Security/negative testing
Password displayed as plain text	High	UI/security testing
Logout does not end session	High	Session testing

Incorrect error message	Medium	Validation testing
Login is very slow	Medium	Performance testing
Login fails on a browser	Medium	Compatibility testing

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7. Features to Be Tested

The main features and corresponding testing areas are:

Feature	Testing
Username	Valid, invalid, empty, special characters
Password	Valid, invalid, empty, masking
Login	Positive and negative authentication
Validation	Empty/invalid input handling
Error Messages	Accuracy and usability
Session	Login, logout, expiry/back navigation
Security	Unauthorized access and credential handling
UI	Alignment, labels, buttons, readability
Compatibility	Chrome, Firefox, Edge
Performance	Login response baseline

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8. Features Not to Be Tested

The following are excluded from this test cycle:

- Registration functionality
- Profile functionality
- Book search and management
- Book purchase functionality

- Complete backend/database testing
- Full penetration/security audit

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9. Test Approach / Strategy

Testing will follow a **risk-based and requirement-based approach**.

Step 1 – Requirement Analysis

Understand login requirements and identify:

- Functional requirements
- Validation rules
- Security expectations
- UI expectations
- Session behavior

Step 2 – Test Design

Test cases will be created using:

- **Equivalence Partitioning** – divide inputs into valid/invalid groups.
- **Boundary Value Analysis** – test values around input limits.
- **Decision Table Testing** – test combinations of username/password conditions.
- **State Transition Testing** – test Logged Out → Logged In → Logged Out.
- **Use Case Testing** – test complete user login flow.
- **Error Guessing** – test likely failure conditions.
- **Exploratory Testing** – investigate unexpected behavior.

Step 3 – Test Execution

1. Perform Smoke Testing.
2. Reject/block the build if critical login functionality is unavailable.
3. Execute functional and negative test cases.
4. Perform security, session, UI and compatibility testing.
5. Report defects.
6. Retest fixes.
7. Perform regression testing.
8. Prepare final test report.

10. Example Test Scenarios

ID	Scenario	Expected Result
TS-01	Login with valid credentials	User successfully logs in
TS-02	Invalid username + valid password	Login rejected
TS-03	Valid username + invalid password	Login rejected
TS-04	Invalid username + invalid password	Login rejected
TS-05	Empty username	Validation/error displayed
TS-06	Empty password	Validation/error displayed
TS-07	Both fields empty	Validation/error displayed
TS-08	Password field input	Password is masked
TS-09	Logout	User is logged out
TS-10	Access protected page after logout	Access is denied/redirected
TS-11	Browser back after logout	Logged-out state remains
TS-12	Special-character input	Application handles input safely

11. Example Decision Table

Username	Password	Expected Result
Valid	Valid	Login successful
Valid	Invalid	Login rejected
Invalid	Valid	Login rejected
Invalid	Invalid	Login rejected
Empty	Valid	Validation/error

Valid	Empty	Validation/error
Empty	Empty	Validation/error

This ensures important combinations are not missed.

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12. Pass/Fail Criteria

Pass

A test passes when:

- Actual result matches expected result.
- Required validation works correctly.
- No unexpected behavior occurs.
- Evidence supports the result where required.

Fail

A test fails when:

- Actual result differs from expected result.
- Required functionality does not work.
- Incorrect validation/error occurs.
- Unauthorized behavior is observed.

Example:
Expected: Invalid password should be rejected.
Actual: User successfully logs in.
→ FAIL + Security/Critical defect

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13. Suspension and Resumption Criteria

Testing will be suspended if:

- Login page is unavailable.
- Login button does not function.
- Application crashes.
- Backend/authentication service is unavailable.
- Valid test credentials/data are unavailable.
- A critical blocker prevents meaningful testing.

Testing will resume when:

- Blocking issue is fixed.
- Environment is available.
- Required test data is available.
- Smoke testing passes.

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14. Defect/Bug Reporting and Management

Every deviation between expected and actual behavior will be evaluated and reported when it is confirmed as a defect.

Defect Lifecycle

New → Assigned → Open → Fixed → Retest → Verified → Closed

If the fix fails:

Retest Failed → Reopened → Fixed → Retest

Required Bug Report Fields

Field	Example
Bug ID	BUG-LOGIN-001
Title	Login succeeds with invalid password
Module	Login
Environment	Chrome / Windows
Preconditions	Valid username available

Steps	Enter valid username → invalid password → Login
Expected	Login should be rejected
Actual	User is logged in
Severity	Critical
Priority	P1
Evidence	Screenshot/video
Status	New
Assignee	Developer
Retest Result	Pass/Fail

Severity

- **Critical:** Major security/functionality failure.
- **High:** Major functionality unavailable.
- **Medium:** Functionality affected but workaround exists.
- **Low:** Minor UI/cosmetic issue.

Priority

- **P1:** Fix immediately.
- **P2:** Fix before release if possible.
- **P3:** Can be fixed later.

Example:
Login accepts incorrect passwords → **Critical Severity + P1 Priority**.

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15. Retesting and Regression Testing

Retesting

Retesting verifies that a reported defect has been fixed.

Example:
BUG-LOGIN-001: Invalid password allows login.
After the developer fixes it, execute the same test again.

Regression Testing

Regression testing verifies that a fix has not broken existing functionality.

Example:
After fixing password validation, also test:

- Valid login
- Invalid login
- Empty password
- Logout
- Session behavior

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16. Test Metrics and Reporting

The following metrics will be recorded:

Metric	Formula
Execution %	Executed ÷ Planned × 100
Pass %	Passed ÷ Executed × 100
Fail %	Failed ÷ Executed × 100
Defect Closure %	Closed ÷ Total Defects × 100
Retest Pass %	Passed Retests ÷ Total Retests × 100

Example

If 50 test cases are planned:

- Executed = 50
- Passed = 45
- Failed = 5

- Defects = 5

Execution = **100%**
Pass Rate = **90%**
Fail Rate = **10%**

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17. Test Deliverables

Deliverable	Purpose
Test Plan	Defines testing scope, approach and resources
Test Scenarios	Identifies functionality to test
Test Cases	Provides detailed test steps
Test Data	Provides required inputs
Bug Reports	Documents defects
Test Execution Report	Records test results
Retest Report	Records verification of fixes
Test Summary Report	Provides final testing status

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18. Requirements Traceability

Requirements will be mapped to test scenarios and test cases.

Requirement → Scenario → Test Case → Result → Defect

Example:

REQ-LOGIN-001: User must log in with valid credentials.
↓
TS-LOGIN-001: Login using valid credentials.
↓

TC-LOGIN-001: Enter valid username/password.
↓
PASS/FAIL
↓
BUG-LOGIN-001 if failed.

This ensures that no important requirement is left untested.

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19. Test Environment and Configuration

Hardware

- Desktop/Laptop
- Mobile/Tablet where applicable

Operating Systems

- Windows
- Android/mobile environment where applicable

Browsers

- Google Chrome
- Mozilla Firefox
- Microsoft Edge

Other Conditions

- Normal network
- Slow/intermittent network
- Browser DevTools
- Screenshot/recording tools
- Test management/bug tracking tool

Configuration to Record

- Application URL
- Application build/version
- OS/version

- Browser/version
- Device
- Screen resolution
- Network condition
- Test data
- Execution date

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20. Staffing and Training Needs

Required Roles

- QA/Test Engineer
- QA Lead
- Developer
- Project Manager
- Business Stakeholder

Training

Testers should understand:

- Application requirements
- Login functionality
- Test design techniques
- Defect reporting
- JIRA/Zoho or selected tracking tool
- Basic security/session testing
- Browser DevTools

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21. Roles and Responsibilities

Role	Responsibility
QA/Test Engineer	Design, execute, report, retest and regression test

QA Lead	Review plan/cases and monitor quality
Developer	Analyze and fix defects
Project Manager	Schedule and resource management
Business Stakeholder	Clarify requirements and acceptance
QA Team	Prepare final test summary

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22. Time Estimation

Activity	Estimated Effort
Requirement Analysis	2 hrs
Test Plan	3 hrs
Scenario Creation	2 hrs
Test Case Creation	4 hrs
Test Case Review	1 hr
Test Data/Environment	1 hr
Smoke Testing	1 hr
Functional Testing	5 hrs
Negative/Boundary Testing	3 hrs
Security/Session Testing	3 hrs
Browser/UI Testing	2 hrs
Defect Reporting	1 hr
Retesting	2 hrs
Regression Testing	2 hrs
Test Summary Report	2 hrs

Total Estimated Effort	34 hrs

Note: Actual effort may change depending on defects, application changes, environment availability, and additional testing cycles.

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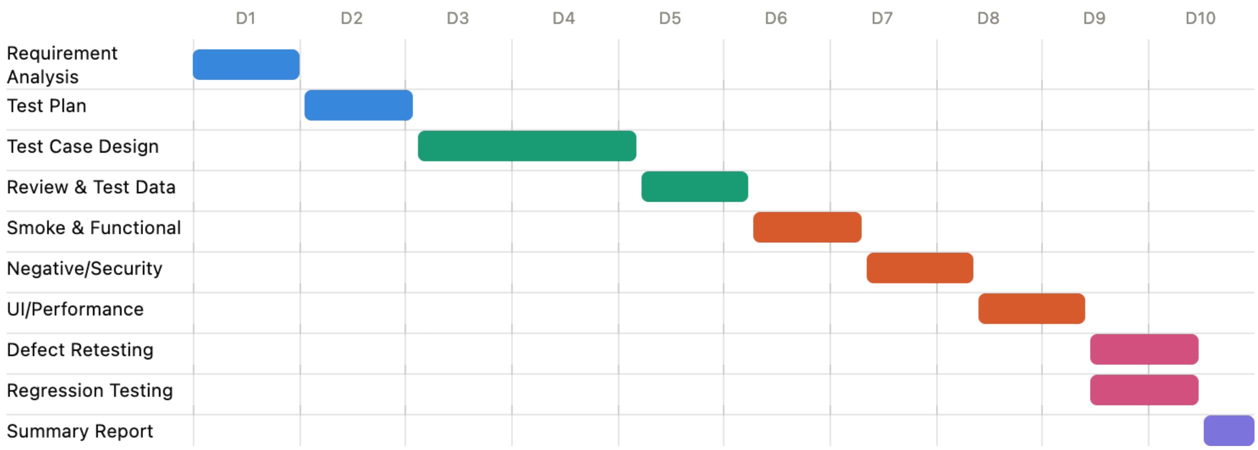
23. Test Schedule



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24. Planning Risks and Contingencies

These risks affect the **testing process**, rather than the software itself.

Planning Risk	Contingency
Application unavailable	Perform documentation/design work
Limited testing time	Prioritize critical/risk-based tests
Environment unavailable	Use alternate environment
Developer fixes delayed	Continue unaffected test areas
Requirements unclear	Contact BA/PM/stakeholder
Browser unavailable	Test supported alternative

Insufficient test data	Prepare additional test data

25. Test Closure

Testing will be considered complete when:

- Planned testing is completed.
- Critical login functionality has passed.
- Critical/High defects are fixed or formally accepted.
- Required retesting is completed.
- Regression testing is completed.
- Test results are documented.
- Outstanding risks are reported.
- Test Summary Report is prepared.
- QA Lead/stakeholder approval is obtained.

Final Test Summary Example

Overall Status: PASS / PASS WITH KNOWN ISSUES / FAIL

The report should contain:

- Planned vs executed tests
- Pass/fail results
- Defect summary
- Severity/priority distribution
- Retest results
- Regression results
- Outstanding risks
- Final recommendation

26. Tools

- **JIRA / Zoho Sprints** – defect management
- **Excel** – test scenarios/test cases
- **Mind Map Tool** – requirement/test analysis

- **Browser DevTools** – debugging and network inspection
- **Screenshot/Recording Tool** – defect evidence
- **Word/Google Docs** – documentation

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27. Approvals

Document	Approval
Test Plan	QA Lead / Project Manager
Test Scenarios	QA Lead
Test Cases	QA Lead / Stakeholder
Test Summary Report	QA Lead / Business Stakeholder

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28. Glossary

Term	Meaning
AUT	Application Under Test
QA	Quality Assurance – prevents quality problems through processes
QC	Quality Control – identifies quality problems through testing/inspection
Test Case	Detailed steps, data and expected result
Test Scenario	High-level functionality to be tested
Defect/Bug	Flaw in the software
Error	Human mistake that can introduce a defect

Failure	Incorrect behavior observed during execution
Risk	Potential future problem
Severity	Impact of a defect
Priority	Urgency of fixing a defect
Regression	Testing existing functionality after changes
Retesting	Testing a previously failed test after its defect is fixed
Smoke Testing	Basic testing to determine whether a build is testable
BVA	Boundary Value Analysis
EP	Equivalence Partitioning
Authentication	Verifying who the user is
Authorization	Determining what the user can access
Session	State maintained after successful login